

# Posterior Estimation via Conjugacy: Beta–Binomial Model

COM3031 Probabilistic Machine Learning

## 1 Model Specification

Suppose a coin is flipped  $n$  times and  $y$  heads are observed. Let  $\theta \in (0, 1)$  denote the probability of obtaining heads.

### Likelihood

Conditioned on  $\theta$ , the number of heads follows a binomial distribution:

$$p(y \mid \theta) = \binom{n}{y} \theta^y (1 - \theta)^{n-y}.$$

### Prior

We place a Beta prior on  $\theta$ :

$$\theta \sim \text{Beta}(\alpha, \beta),$$

with density

$$p(\theta) = \frac{1}{B(\alpha, \beta)} \theta^{\alpha-1} (1 - \theta)^{\beta-1},$$

where  $B(\alpha, \beta)$  is the Beta function.

## 2 Posterior Distribution via Conjugacy

By Bayes' theorem,

$$p(\theta \mid y) \propto p(y \mid \theta) p(\theta).$$

Substituting the likelihood and the prior:

$$p(\theta \mid y) \propto \binom{n}{y} \theta^y (1 - \theta)^{n-y} \cdot \frac{1}{B(\alpha, \beta)} \theta^{\alpha-1} (1 - \theta)^{\beta-1}.$$

Ignoring normalising constants that do not depend on  $\theta$ , we obtain

$$p(\theta \mid y) \propto \theta^{\alpha+y-1} (1 - \theta)^{\beta+(n-y)-1}.$$

## 3 Identification of the Posterior

The resulting expression has the same functional form as a Beta distribution. Therefore, by conjugacy,

$$\theta \mid y \sim \text{Beta}(\alpha_0, \beta_0),$$

where

$$\boxed{\alpha_0 = \alpha + y, \quad \beta_0 = \beta + (n - y).}$$

## Posterior Mean

$$\mathbb{E}[\theta \mid y] = \frac{\alpha_0}{\alpha_0 + \beta_0} = \frac{\alpha + y}{\alpha + \beta + n}.$$

## 4 Interpretation

For a Binomial likelihood with a Beta prior, the posterior distribution is also Beta. Posterior inference follows immediately by updating the parameters via conjugacy, without optimisation.